

COURSE SYLLABUS-Fall 2025

**CHE321 CHEMICAL REACTION
ENGINEERING**

1. Course Title

Prefix: CHE 321, Number: 43300, and Title: Chemical Reaction Engineering

2. Course Credit

3 Credit Hours

3. Placement in Curriculum

Junior year, core chemical engineering course.

4. Prerequisites

CHE 210; and MATH 220; and CHE 301

5. Course Professor and Teaching Assistant

Lead Professor

Name: Meenesh R. Singh

Title: Professor

Phone: 312-413-7673

Email: mrsingh@uic.edu

Teaching Assistant

Name: Gowtham Kumar

Title: Graduate Student

Email: gvadl@uic.edu

6. Lecture and Office Hours

Lecture Hours

Time- Fridays, 9:30 am to 12:15 pm.

Lecture Type- In-Person

Platform- EIB 271

Instructor Office Hours

Time- Fridays, 8:30 am to 09:30 am

Location: EIB 256

TA Office Hours

Time- Fridays, 12:30 pm to 01:00 pm

Location: EIB 271 or Library

7. Course Description

(From the course catalog)- Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors. Non-isothermal reactors.

8. Course Rationale

Chemical Reaction Engineering (CRE) is central and unique to the chemical engineering discipline. The elements of CRE are used in almost all the chemical industries and manufacturing, namely, petrochemicals, biopharmaceuticals, agrochemicals, polymers, fragrances, flavors, food additives, dyes and chemicals, ceramics, explosives and many more. The heart of these manufacturing processes is a chemical reaction, which converts raw materials or precursors to the molecule of interest. The objective of chemical engineers is to maximize the yield and conversion of the final product in the most energy-efficient and cost-effective manner. This course will teach fundamental concepts in CRE, provide step-by-step instructions on the evaluation of the performance of the chemical reactor and its optimization.

9. Course Learning Objectives

Module-1: Chemical Kinetics and Ideal, Isothermal Homogeneous Reactors (Ch-1 to 6)

- Define the rate of chemical reaction. Apply the mole balance equations to a batch reactor (BR), continuous stirred tank reactor (CSTR), plug flow reactor (PFR), and packed bed reactor (PBR).
- Define conversion and space time. Write the mole balances in terms of conversion. Determine reactor sizes (volume, catalyst weight).
- Understand rate laws and the Arrhenius equation. Describe homogeneous, heterogeneous, elementary, nonelementary and reversible reactions.
- Apply reaction stoichiometry to relate molar and volumetric flow rates of a species with the conversion.
- Describe the CRE algorithm to solve chemical reaction engineering problems. Apply this algorithm to design isothermal reactors.

Module-2: Experimental Analysis of Reaction Rates (Ch-7)

- Determine reaction order and rates from the experimental data obtained from batch or flow reactors.
- Understand integral and differential methods to identify rate parameters.

Module-3: Ideal, Non-Isothermal Reactors (Ch-8, 11, 12, 13)

- Describe yield and selectivity. Apply CRE algorithm to design the reactor with multiple reactions for maximal selectivity.
- Modify CRE algorithm to design non-isothermal reactors (adiabatic and non-adiabatic) and optimize the reactor staging.
- Analyze multiple steady states and optimal operation of reactors.

Module-4: Catalytic Reactions and Heterogeneous Reactors (Ch-10, 14, 15)

- Define a catalyst, derive a catalytic reaction mechanism and rate-limiting step
- Obtain rate expressions from quasi-equilibrium hypothesis.
- Understand catalyst deactivation mechanism
- Analyze heterogeneous catalytic reactors – CSTR and PBR

Module-5: Non-ideal Reactors (Ch-16, 17, 18)

- Define residence time distribution (RTD) and obtain RTD functions for ideal and non-ideal reactors
- Predict conversion from RTD functions
- Evaluate non-ideal reactor models.

10. Materials

Required Text

H Scott, Fogler. *Elements of chemical reaction engineering*. 5th Edition, Prentice Hall Professional, 2006.

Read it online here <https://www.oreilly.com/library/view/elements-of-chemical/9780135486252/?ar>
You will need access to O'Reilly to read this book.

Recommended (Optional) Text

Finlayson, Bruce A. *Introduction to chemical engineering computing*. Wiley-Interscience, 2006.

Other Resources

<http://umich.edu/~elements/5e/index.html>

11. Course Outline and Expectations

Weekly schedule of topics to be covered in lectures are listed below

Week	Module	Topics to be covered	Textbook Chapters
Week 1	Module 1	General mole balance, examples of reactor, conversion, design equations for reactors	Ch-1,2

Week 2	Module 1	Sizing of reactors, reactors in series, space time and space velocity, relative rates and reaction invariant, rate laws, Elementary rate laws, non-elementary rate laws, reversible reaction, rate constant, rate order, Arrhenius plot	Ch-2,3
Week 3	Module 1	Problem Solving Ch-1, 2, and 3. Stoichiometry, concentration- conversion relationship, equilibrium conversion. Problem Solving Ch-4	Ch-1,2,3, 4
Week 4	Module 1	CRE algorithm for isothermal reactors, design of BR, CSTR, and tubular reactors, pressure drop in reactors and its effect	Ch-5
Week 5	Module 1	CRE algorithm in terms of moles, membrane reactor design, unsteady-state operation, and semi-batch operation	Ch-6
Week 6	Module 1	Problem Solving Ch-6 and 7. Review of Module 1	Ch-6
Week 7	Module 2	Data analysis algorithm, method of excess, integral method, differential method, non-linear regression, differential reactor	Ch-7
Week 8	Module 2	Multiple reactions , selectivity, yield, CRE algorithm for multiple reactions, parallel reaction, series reaction, complex reaction	Ch-8
Week 9	Module 3	Energy balance, adiabatic operation, reactor staging, optimal feed temperature	Ch-11
Week 10	Module 3	Steady-state tubular reactor, CRE algorithm with heat effects, multiple steady states, non-isothermal multiple reaction with spatial variation	Ch-12
Week 11	Module 3	Unsteady state operation, adiabatic BR, CSTR startup.	Ch-13
Week 12	Module 4	Catalyst, steps in catalytic reaction, synthesizing rate law, rate-limiting step, PSSH, estimating rate law, catalyst deactivation	Ch-10
Week 13	Module 4	Review of diffusion, Fick's law, diffusion and reaction in homogeneous and heterogeneous systems, effectiveness factor, Thiele modulus, falsified kinetics, multiphase reactor	Ch-10,14
Week 14	Module 5	RTD, measurement of RTD, RTD of ideal reactors	Ch-16

Week 15	Module 5	RTD models for non-ideal reactors-segregation, maximum mixedness, tank in series, and dispersion models	Ch-17, 18
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Participation/Attendance

Active participation in lectures is recommended. Class participation has 10% weightage to overall grade evaluation.

Academic Honesty

Students in CHE 321 are continuing on a program of study to become chemical engineers. Engineers are expected to uphold a strict code of ethics in which they: (1) build their professional reputations on the merits of their services and (2) act in such a manner as to uphold and enhance the honor, integrity, and dignity of the engineering profession. In addition to these broad ethical guidelines, students in CHE 321 will be expected to be honest and refrain from cheating on exams, and homework assignments (you should not turn in a homework that you did not work yourself).

Students caught cheating will be prosecuted to the maximum extent possible under UIC guidelines. Personally, I take all accusations of academic dishonesty very seriously, and I am asking each you to report any incidents or suspected incidents of cheating to me directly. I will keep your identity confidential.

Accommodations

The University of Illinois at Chicago is committed to maintaining a barrier-free environment so that students with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for access to and/or participation in this course are welcome, but must be registered with the Disability Resource Center (DRC). You may contact DRC at 312-413-2183 (v) or 773-649-4535 (VP/Relay) and consult the following:

<http://drc.uic.edu/guide-to-accommodations>.

If you are not vaccinated, you may request for exemption. More information can be found at <https://today.uic.edu/student-covid-19-vaccination-requirements-and-deadlines>. If you do not have a vaccination exemption, some students can qualify for special accommodation by contacting the [Disability Resource Center](#). If your accommodation is approved, I will make my best effort to provide the best remote instruction.

Course Communication Policy

Send email correspondence with subject "[CHE 321]"

Course Communication Guidelines (Netiquette)

Netiquette is a set of rules for behaving properly online. Much of our communication in

this course will take place in the forums and through email. Here are some guidelines for online communication in this course:

- Be sensitive to different cultural and linguistic backgrounds, as well as different political and religious beliefs.
- Use good taste when composing your responses. Swearing and profanity should be avoided. Also consider that slang terms can be misunderstood or misinterpreted.
- Don't use all capital letters when composing your responses. This can be considered "shouting" on the Internet and is regarded as impolite or aggressive. It can also be stressful on the eye when trying to read your message.
- Be respectful of others' views and opinions. Avoid "flaming" (publicly attacking or insulting) others.
- Be careful when using acronyms. If you use an acronym it is best to spell out its meaning first, then put the acronym in parentheses afterward, for example: Frequently Asked Questions (FAQs). After that you can use the acronym freely throughout your message.
- Use good grammar and spelling, and avoid using text messaging shortcuts.
- In emails, always identify yourself and what class and section you are in. It is good practice to put your course and section in the subject line. This helps your instructor identify course-related emails.

12. Technology Requirements

Computer/Technology Requirements

Online students will need regular access to a personal computer that runs on a broadband Internet connection.

Blackboard Learning Management System

For all technical questions about Blackboard, email ACCC-Learning Technology Solutions at LTS@uic.edu

Web Conferences/Synchronous sessions

All lectures will be delivered in-person. If needed, we will use Blackboard Collaborate Ultra for recording of lecture- <https://uic.blackboard.com/webapps/collab->

13. Grading and Evaluation

Final Grade Calculation

Assessments	Date	Percentages
Weekly Assignments	Fridays	20%
Class Participation	Fridays	5%
Surprise Quizzes	Fridays	5%
Exam 1 (in-person)	10/03/2025	20%
Exam 2 (in-person)	11/07/2025	20%
Final Exam (in-person)	12/12/2025	30%
Total Percentage For the Course		100%

Assignments: Assignments will be posted on Monday and will be due on next Monday. Any assignment you turn in is to be a product of your own efforts and your own understanding. If you do not understand what you are writing – DON'T WRITE IT. Anyone who copies the solution from another student or a solution manual is guilty of academic dishonesty. Anyone suspected of copying may be given an announced quiz to determine if their work was their own. Even if you "worked with someone", you must do it yourself.

Grading

In determining the final course grade, the following scale is used:

>85% = A

75% - 85% = B

60% - 75% = C

< 60% = F

Late Work Policy

The only late work that will be considered are those instances where the student has communicated with their instructor regarding unavoidable circumstances, such as reporting an illness and submitting a doctor's note to their instructor. All other late work

within 24 hours from deadline will receive 50% penalty, and submission received after 24 hrs will not be graded.

Final Exams

Although the demands of some disciplines may require specific scheduling of end of term projects or final exams, in general, final examinations, if they are to be given, should only be administered during the sixteenth week of the semester (eighth week of the summer term). They may not be given earlier. The dates and times of the exams are scheduled by the Office of Classroom Scheduling and are arranged so that a student, ideally, will have no more than two exams in a day. Any student having more than two final examinations scheduled in one day is entitled to rescheduling. No formal instruction of any kind may be given during the final examination period. Faculty are expected teach through the last week of the semester and should not administer major exams during the last two weeks of the term. Variance from these guidelines must meet approval from the unit /department head. For a schedule of exams and exam policies see:

http://www.uic.edu/depts/oar/current_students/calendars/final_exam_schedule.html

GRIEVANCE PROCEDURES

UIC is committed to the most fundamental principles of academic freedom, equality of opportunity, and human dignity involving students and employees. Freedom from discrimination is a foundation for all decision making at UIC. Students are encouraged to study the University's "[Nondiscrimination Statement](#)". Students are also urged to read the document "[Public Formal Grievance Procedures](#)". Information on these policies and procedures is available on the University web pages of the Office of Access and Equity: <http://oae.uic.edu/>.

14. Course Evaluation

Student evaluations of teaching play a fundamental role in improving course content, format, and delivery (teaching) at UIC. The Office for Faculty Affairs offers all Colleges and Departments the opportunity to participate in an online course evaluation system.

Students receive an email invitation in their 'uic.edu' inbox with the following title in the subject line: "UIC Student Evaluation of Teaching [Course Name] [Instructor Name] [Semester, Year]." The body of the email will reiterate the course name, instructor name, and semester. It will contain a link and a unique student password for the online evaluation for that course. Students will need an electronic device with Internet access to complete the evaluation online. **Submitted course evaluations cannot be removed from the system** so it is vital that students pay attention to the *instructor name* and *course name* when completing their evaluations. **Students must complete the online evaluations before 12**

am on the first day of finals. No exceptions are made if the evaluation is not submitted before 12 am on the first day of finals.

For more information about the program and timelines for when the system is open to students to complete the evaluations, please visit:

<https://faculty.uic.edu/development/teachingandlearning/evaluations/>

15. Course Topics

Modules	Title
Module 1	Chemical Kinetics and Ideal, Isothermal Homogeneous Reactors (Ch-1 to 6)
Module 2	Experimental Analysis of Reaction Rates (Ch-7)
Module 3	Ideal, Non-Isothermal Reactors (Ch-8, 11, 12, 13)
Module 4	Catalytic Reactions and Heterogeneous Reactors (Ch-10, 14, 15)
Module 5	Non-ideal Reactors (Ch-16, 17, 18)

16. UIC Resources

If you find yourself having difficulty with the course material or any other difficulties in your student life, don't hesitate to ask for help! Come to me, or if it is about an issue beyond this class, please contact your college advisors, or get help from any number of other support services on campus. You can get a referral to the right place, or help on the spot, from a concerned advisor in the Undergraduate Success Center (USC) at usc@uic.edu.

See also:

The Writing Center offers friendly and supportive tutors who can help you with reading and writing in any of your courses, not just English. Tutors are ready to help other writing as well, such as job applications, personal statements, and resumes. The tutor and you will work together to decide how to improve your writing. If you have not started your assignment, that is OK. A tutor can help you brainstorm or make an outline. Tutors understand that you might be using the Writing Center for the first time. They are ready to guide you through your first session. You can choose to work with a tutor in real time using chat and a white board, or submit up to 5 pages of text and receive written feedback within 48 hours. To schedule an appointment, visit <https://uic.mywconline.com/>. For more information, visit the Writing Center website at <https://writingcenter.uic.edu/>.

The Math and Science Learning Center, located in the Science and Engineering South Building (SES) at 845 W. Taylor St. 3rd Floor, Room 247, is a meeting place for students in Math, Biological Sciences, Chemistry, Earth and Environmental Sciences, and Physics. At the MSLC, students can meet with graduate teaching assistants for tutoring in 100-level courses, arrange informal group study sessions with other students, or meet up with friends to attend one of the workshops, seminars, or other activities sponsored by the SLC during the semester. Visit the website at <https://mslc.uic.edu/>, call 312-355-4900 or email at mslc@uic.edu.

The UIC Library

The library is located both on east and west campus, provides access to resources, study rooms, and research support both online via chat and in person. At Daley Library on the east side of campus, stop by the reference desk in the IDEA Commons, or make an appointment for research help on either side of campus. Learn more about library policies at <http://library.uic.edu/>. To find research materials in specific subject areas view the Research Guides at <http://researchguides.uic.edu/>.

The Academic Center for Excellence can help if you feel you need more individualized instruction in reading and/or writing, study skills, time management, etc. Phone: (312) 413-0031.

Counseling Services are available for all UIC students. You may seek free and confidential services from the Counseling Center www.counseling.uic.edu. The Counseling Center is located in the Student Services Building; you may contact them at (312) 996-3490. In addition to offering counseling services, the Counseling Center also operates the InTouch Crisis Hotline from 6:00 p.m.-10:30 p.m. They offer support and referrals to callers, as well as telephone crisis interventions; please call (312) 996-5535.

Campus Advocacy Network

Under the Title IX law you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim-services and advocacy contact UIC's Campus Advocacy Network at 312-413-1025 or visit <http://can.uic.edu/>. To make a report to UIC's Title IX office, email TitleIX@uic.edu or (312) 996-5657.

CAMPUS SECURITY

As a UIC student, you've chosen to live in one of the nation's largest cities. But, as at any university, crime is a reality. At UIC, we are strongly committed to our public safety programs, and we encourage students to be proactive in learning what programs and services are available in case of an emergency. You are DISCOURAGED from staying in university buildings alone, including lab rooms, after hours and are ENCOURAGED to use the POLICE/STUDENT patrol escort if you are uncomfortable traveling anywhere on

campus. You may request an escort to accompany you to your campus destination on foot by calling 312-996-2830, and between 11:00 pm and 7:00 am you can dial the Red Car service (312-996-6800) if you are alone and need to leave the building. Through Red Car, the university has established a safe evening transportation service for university employees, students, visitors, and other authorized individuals. The car travels between university facilities within the following general boundaries: Clinton Street on the east; Western Avenue on the west; Jackson Boulevard on the north; and, 16th on the south. This service is available only to individuals possessing a valid UIC i-card. The i-card is required to ensure the safety of the driver and other passengers. Consult the following for more information: <http://www.uic.edu/uic/studentlife/campus/safety.shtml>

Also you can subscribe your cell phone to receive text message alerts. An immediate SMS text alert will be sent in case of a serious crime in progress, a weather emergency, or other urgent situation. (https://accountportal.uic.edu/auth/manage?dispatch=set_sms_panel). Finally, by dialing 5-5555 from a campus phone, you can summon Police or Fire for any on-campus emergency. You may also set up the complete number, 1-312-355-5555, on speed-dial on your cell phone. For more information contact: <http://www.uic.edu/uic/studentlife/campus/emergency-information.shtml>

Emergency Response Recommendations

You may wish to include the following section on emergency response systems at UIC:

The emergency response guide can be found at the following website: <https://ready.uic.edu/>. Please review and acquaint yourself with the guide and recommendations for various emergency situations.